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Analysis of interactions between regulated distribution transformers (RDT) and local voltage control Q(V) by means of the harmonic balance method

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ABSTRACT

The increase of decentralized power generation in low voltage grids imposes new challenges on the conventional operating strategies for electrical grids, as power transmission used to predominantly pass from high voltage levels to low voltage levels. Distributed generation in low voltage grids lead to unprecedented voltage limit violations, which can efficiently be solved by implementing new control systems e.g. regulated distribution transformers (RDT) or local voltage control (e.g. Q(V)). Both correspond to closed-loop control systems and thus, are prone to stability issues and possible interactions with other control elements. Furthermore, the nonlinear characteristics of RDTs complicate the systems' stability analysis. This paper introduces the harmonic balance method as an approach to characterize the nonlinear behavior of an RDT and assess the system stability as well as possible interactions with other linear controllers in the electrical grid.

1. Introduction

Due to the increase of decentralized power generation in European low voltage grids, conventional operating strategies of electrical power grids have to be modified. Decentralized voltage control offers a efficient solution to stay within the voltage limits. Control systems like the local voltage control Q(V) and RDT correspond to a closed-loop control structure and thus, they are prone to stability issues. The stability of Q(V) controllers has been previously researched for simplified controllers [1,2] as well as with respect to the nonlinear behavior [3,4]. Voltage control systems can interact with other voltage controllers like regulated distribution transformer (RDT) as shown in [5], where a oscillation due to interaction of a transformer and a switched capacitor is portrayed. Examples for tap changes induced by the reactive power control of distributed energy resources are given in [6] and [7]. The study [8] analyzes possible opposing reactions of Q(V)-control and RDTs for electrical grids with regards to unfavorable steady state operating points of the RDT and the Q(V)-control. One of such operating points could be a stable oscillation leading to periodical tap position changes, which increase stress on the RDT. While the previous publications analyzed time domain simulations in order to identify interactions, an in-depth analysis using control theory still has to be conducted to identify systemic interactions. One approach to analyze oscillations of nonlinear elements is the harmonic balance method, which also has previously been used to predict harmonics in power systems [9]. This paper aims to introduce the harmonic balance method as an approach to identify possible oscillations of an RDT with other voltage controllers in the grid. Finally, the method is applied to an RDT and two exemplary Q(V)-control systems for decentralized generators.

2. Harmonic balance method

The interactions between a non-linear and a linear control, as depicted in Fig. 1, can be described by using the Harmonic Balance (HB) method [10]. It allows to analyze possible operating points, which lead to stable oscillations between the two elements within the closed-loop control system.

The basic requirements for the utilization of the Harmonic Balance method are a point-symmetric structure of the non-linear characteristic $u = f(e, \dot{e})$ with the error value e and the input value of the linear system u. Furthermore, the linear control $G(s)$ has to show a sufficient low-pass filter characteristic for the control variable y. In order to evaluate the HB of the system without external influences, the system has to be in steady state. The reference signal r is assumed to be zero [11].

2.1. Harmonic balance theory

A stable oscillation of the control error value e is defined by Eq. (1) with the amplitude A and the angular velocity ω, where the control

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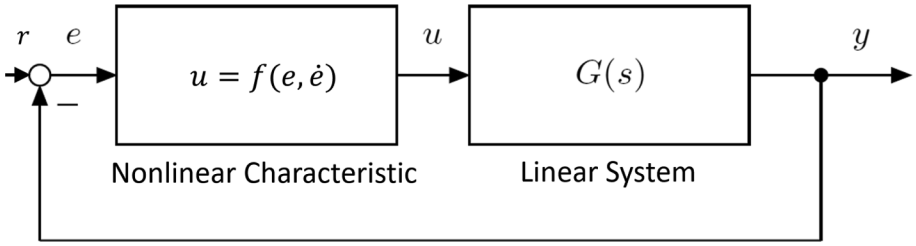


Fig. 1. Non-linear control loop for the Harmonic Balance [10].

variable y is shifted by π compared to the e due to the non-linear function and the transfer function as described in (2). [10]

$$e(t) = A\sin(\omega t) = Re\{-jAe^{j\omega t}\} \tag{1}$$

$$e(t) = A\sin(\omega t) = -A\sin(\omega t - \pi) = -y(t) \tag{2}$$

The transformation of the error value e by the non-linear function $f(e, \dot{e})$ to the input variable u can be described by a Fourier series (3). Due to the point-symmetric characteristic of the non-linear function, the steady component b_0 can be assumed as zero. Additionally, all amplitudes of the higher harmonics a_i, $b_i for i > 1$ can be discarded because of the required low-pass filter characteristic of the linear system. This leads to the equation for the fundamental wave in Eq. (4) [11].

$$\begin{aligned} u(t) &= b_0 + \textstyle\sum_{i=1}^{\infty} a_i\sin(i\omega t) + b_i\cos(i\omega t) \\ b_0(A) &= 0 a_1,\ b_1 \gg a_i,\ b_i i = 2, 3, \ldots \end{aligned} \tag{3}$$

$$\begin{aligned} u(t) &= a_1\sin(\omega t) + b_1\cos(\omega t) \\ &= Re\{(b_1 - ja_1)e^{j\omega t}\} \end{aligned} \tag{4}$$

Thus, y is calculated as

$$y(t) = Re\{G(j\omega)(b_1 - ja_1)e^{j\omega t}\} \tag{5}$$

The signal e is transformed to a sine wave with the same frequency but with a shift in amplitude and angle. This behaviour is comparable to a linear transfer function and can be described with (2) using (4) and (5) as (6) [11].

$$\begin{aligned} -jAe^{j\omega t} &= -G(j\omega)(b_1 - ja_1)e^{j\omega t} \\ A &= -G(j\omega)(a_1 + jb_1) \\ -1 &= G(j\omega)\tfrac{a_1 + jb_1}{A} \end{aligned} \tag{6}$$

The behaviour of the non-linear element is defined by the describing function $N(A)$ in (7), which only depends on the amplitude A of the oscillation and is not time-dependent [10].

$$N(A) = \frac{a_1}{A} + j\frac{b_1}{A} \tag{7}$$

This leads to the oscillation criteria (8), which defines the steady state operation between the linear and the non-linear control. $N_I(A)$ is the negative inverse function of the describing function $N(A)$ [10].

$$\begin{aligned} N(A)\cdot G(j\omega) &= -1 \\ G(j\omega) = -N(A)^{-1} &= N_I(A) \end{aligned} \tag{8}$$

If a stable oscillation exists, it's angular velocity ω and amplitude A can be calculated by for solving the real and imaginary part of (8).

2.2. Describing function $N(A)$

The non-linear function f can be further split for characteristics with hysteresis into (9), where f_+ stand for positive and f_- for negative gradients of e [10,11].

$$u = f(e, \dot{e}) = \begin{cases} f_+(e) \text{for} \dot{e} > 0 \\ f_-(e) \text{for} \dot{e} < 0 \end{cases} \tag{9}$$

The describing function can be determined by analyzing the fundamental wave of the Fourier series of the non-linear characteristic, which is composed of a real part $Re\{N(A)\}$ and an imaginary part $Im\{N(A)\}$ defined by Eqs. (10)–(12).

$$\begin{aligned} N(A) &= Re\{N(A)\} + jIm\{N(A)\} \\ v &= \omega t,\ 0 \le v \le 2\pi \end{aligned} \tag{10}$$

$$Re\{N(A)\} = Re = \frac{1}{\pi A}\int_0^{2\pi} f(e, \dot{e})\sin(v)dv \tag{11}$$

$$Im\{N(A)\} = Im = \frac{1}{\pi A}\int_0^{2\pi} f(e, \dot{e})\cos(v)dv \tag{12}$$

In the following, the method for the determination of the describing function is derived for the exemplary characteristic in Fig. 2 on the left.

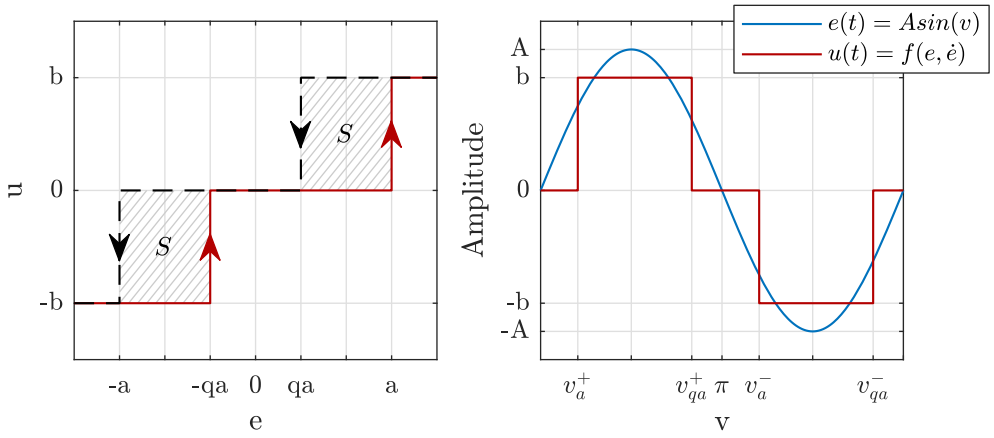


Fig. 2. Exemplary characteristic (left) for the harmonic balance method (red solid line: $f_+(e)$ for $\dot{e} > 0$, black dotted line: $f_-(e)$ for $\dot{e} < 0$) and the respective fundamental wave $e(t)$ and output signal $u(t)$ with the respective switching points (right) [11]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

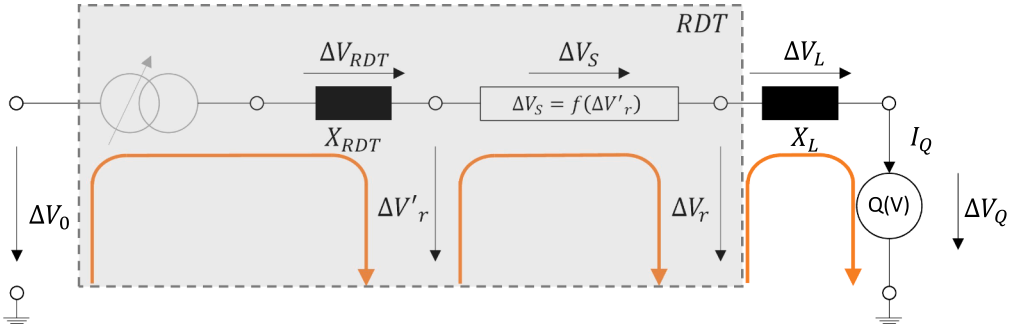


Fig. 3. Simplified electrical model of the grid.

The characteristic shows a varying behavior depending on the rate of change of e (hysteresis) and is defined by the switching points a and qa. It encompasses the area S and mathematically be described by Eqs. (13) and (14) [11].

$$\text{for } \dot{e} > 0 \begin{cases} u = -b & \text{for } e \le -qa \\ u = 0 & \text{for } -qa < e < a \\ u = b & \text{for } e \ge a \end{cases} \tag{13}$$

$$\text{for } \dot{e} < 0 \begin{cases} u = -b & \text{for } e \le -a \\ u = 0 & \text{for } -a < e < qa \\ u = b & \text{for } e \ge qa \end{cases} \tag{14}$$

In order to reach a stable oscillation in terms of a Harmonic Balance, the amplitude of $e(t)$ has to lead to discontinuous behavior on the output of the non-linear characteristic $u(t)$. The instances, at which this occurs, are called switching points v_s [11]

$$e(v_s) = A\sin(v_s)$$

Thus, (15) are the resulting switching points for the positive half cycle of the exemplary non-linear characteristic with an amplitude A greater than the minimal amplitude $A_{min} = a$, as shown in Fig. 2 on the right. The respective points v_a^- and v_{qa}^- for the negative half cycle are shifted by π.

$$\begin{aligned} v_a^+ &= \arcsin\left(\frac{a}{A}\right) \text{with} 0 \le v_a^+ \le \frac{\pi}{2} \\ v_{qa}^+ &= \frac{\pi}{2} + \arccos\left(\frac{qa}{A}\right) \text{with} \frac{\pi}{2} \le v_{qa}^+ < \pi \end{aligned} \tag{15}$$

Due to the point-symmetry of the characteristic, it is sufficient to consider the first half-cycle of the fundamental wave. Therefore, the real part of $N(A)$ is defined by (11) for each piecewise continuous section of (16) as (17) [11].

$$f(e, \dot{e}) = u(v) = \begin{cases} 0 & \text{for } 0 \le v < v_a^+ \\ b & \text{for } v_a^+ \le v < v_{qa}^+ \\ 0 & \text{for } v_{qa}^+ \le v < 2\pi \end{cases} \tag{16}$$

$$\begin{aligned} Re\{N(A)\} &= \frac{2}{\pi A}\int_{v_a^+}^{v_{qa}^+} b\sin(v)dv \\ &= \frac{2b}{\pi A}\left(\sqrt{1-\left(\frac{a}{A}\right)^2} - \sqrt{1-\left(\frac{qa}{A}\right)^2}\right) \end{aligned} \tag{17}$$

The imaginary part of the describing function can be calculated using the same method applied on Eq. (12) and is shown in (18).

$$Im\{N(A)\} = \frac{2}{\pi A}\int_{v_a^+}^{v_{qa}^+} b\cos(v)dv = -\frac{4ab}{\pi A^2} \tag{18}$$

2.3. Stability of HB

The stability of an oscillation according to the HB can be determined by analytically solving the oscillation criteria (8). However, the evaluation is increasingly harder for complicated transfer and describing functions. Therefore, a graphical analysis is advantageous. This is done by plotting the graph of $N_I(A)$ as a function of A and the locus of $G(s)$ as a function of ω in the complex plane. The intersections of both graphs indicate possible operating points. Depending on the direction of graphs relative to each other, the operation point is stable or unstable. If an intersection of $N_I(A)$ with the locus occurs from the left, while moving on the graph of $N_I(A)$ from $A = A_{min}$ to $A \to \infty$, the resulting operating point is stable. Respectively, an intersection from the right leads to an unstable operation point (see Fig. 8) [11].

3. General RDT model

The models used for the numerical validation of the analysis according to the harmonic balance method are simplified electrical grids, a detailed RDT model and Q(V)-controller. All voltages refer to the deviations from the nominal RMS-values (50 Hz) and the frequency of any oscillations shown are therefore to be interpreted as complementary sidebands of an impure beat and not as a superimposed frequency on 50 Hz [12]. Additionally, voltages are in the per unit system, normalized with $S_{base} = 100$ kVA and $V_{base} = 400\ \text{V}/\sqrt{3}$. The oscillations being investigated by the Harmonic Balance originate from the (time-dependent) tap changing voltage control of the RDT. Since the RDT is being operated below its rated apparent power, passive non-linearities (i.e. saturation of the iron core) are not relevant.

3.1. Electrical model

The electrical grid is modeled as a simplified single-phase network and is comprised of an RDT, a simplified line and a Q(V)-controller at the end of the line as depicted in Fig. 3. The primary side of the RDT is connected to an external grid with the voltage V_0. The RDT consists of its reactance X_{RDT} and a variable element, which defines the voltage difference of the tap position ΔV_S as a function of the internal RDT-voltage $\Delta V'_r$. The voltage ΔV_r is the voltage on the secondary side of the RDT. The simplified line is modeled as a reactance X_L. The influence of the calculated oscillations of the harmonic balance on X_L is neglected since the occurring frequencies are complementary sidebands to 50 Hz (impure beat) and the effect on the reactance is therefore marginal [12]. The Q(V)-control as a controlled is modeled as a current source with the reactive current output I_Q.

3.2. RDT-model without time delay

The relevant Eqs. (19)–(21) can be defined directly from Fig. 3. For an initial evaluation of the model, the time delay of the RDT's switching actions is neglected. Furthermore, the magnitudes of the positive and negative tap positions are identical. The current I_Q is approximated by the reactive power output Q_{QV} of the Q(V)-control and the nominal voltage V_N.

$$\Delta V'_r = \Delta V_0 - \Delta V_{RDT} \approx \Delta V_0 - \frac{X_{RDT}}{3V_n} Q_{QV} \tag{19}$$

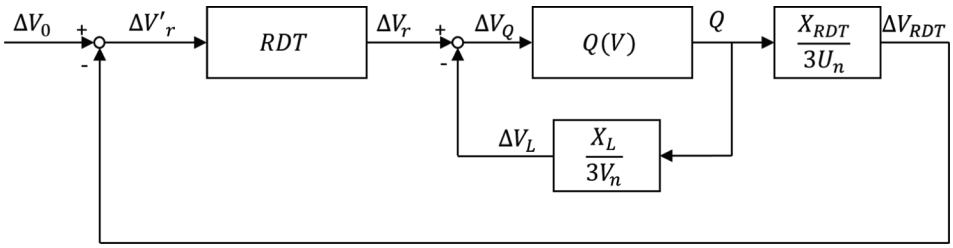


Fig. 4. Control model for an RDT and a Q(V)-control unit in an electrical grid.

$$\Delta V_r = \Delta V'_r - \Delta V_S \tag{20}$$

$$\Delta V_Q = \Delta V_r - \Delta V_L \approx \Delta V_r - \frac{X_L}{3V_n} Q_{QV} \tag{21}$$

The control model for an RDT with a Q(V)-control at the end of the line is defined by Fig. 4, where the non-linear behavior of an RDT with two tap positions follows the graph in Fig. 5. The characteristic is discontinuous at ΔV_{max} and ΔV_{min}. Due to the symmetry of the RDT, the boundaries can be simplified to $\Delta V_{max} = -\Delta V_{min}$.

The exact form of the characteristic is defined by the ratio u_v between the voltage boundary ΔV_{max} and the size of the switching steps ΔV_S, as described by (22). The minimal value for this ratio is 0.5, since the tap control of the RDT itself is oscillating for lower values.

$$u_v = \frac{\Delta V_{max}}{\Delta V_S} \tag{22}$$

The describing function $N(A)$ is calculated based on (11) and (12) with

$$\begin{aligned} v_1^+ &= \arcsin\left(\tfrac{\Delta V_{max}}{A}\right) \\ v_2^+ &= \tfrac{\pi}{2} + \arccos\left(\tfrac{\Delta V_S - \Delta V_{max}}{A}\right) \end{aligned} \tag{23}$$

as (24) and (25). This represents the describing function for the characteristic in Fig. 5 on the left and the respective oscillation on the right.

$$\begin{aligned} &Re\{N(A)\}= \\ &= \tfrac{2}{\pi A}\left[\int_0^{v_1^+} A\sin^2(v)dv + \right. \\ &\quad \left. + \int_{v_1^+}^{v_2^+} (A\sin(v) - \Delta V_S)\sin(v)dv + \int_{v_2^+}^{\pi} A\sin^2(v)dv\right] \\ &= -\tfrac{2\Delta V_S(\sqrt{A^2 - \Delta V_{max}^2} + \sqrt{A^2 - (\Delta V_S - \Delta V_{max})^2}) - \pi A^2}{\pi A^2} \end{aligned} \tag{24}$$

$$\begin{aligned} &Im\{N(A)\}= \\ &= \tfrac{2}{\pi A}\left[\int_0^{v_1^+} A\sin(v)\cos(v)dv + \right. \\ &\quad \left. + \int_{v_1^+}^{v_2^+} (A\sin(v) - \Delta V_S)\cos(v)dv + \int_{v_2^+}^{\pi} A\sin(v)\cos(v)dv\right] \\ &= \tfrac{2\Delta V_S(2\Delta V_{max} - \Delta V_S)}{\pi A^2} \end{aligned} \tag{25}$$

The plots in Fig. 6 show the graphs of $N_I(A)$ of (24) and (25) for different u_v for an RDT with one step each in negative and positive direction. All graphs are in the positive imaginary half plane and shrink for increasing u_v. Furthermore all plots go to $N_I \to -1$ for $A \to \infty$, which can be confirmed by calculating the limits of $N_I(A)$ as shown in [13]. The influence of different ΔV_{max} and ΔV_S for the same u_v has shown to be a difference in how fast the graphs reach values close to $N_I(A) = -1$ for increasing A.

By adding additional tap positions n to the RDT, the graphs grow in the positive imaginary half plane for each additional tap position as shown in Fig. 7. The describing function starts at the marker for

$$A_{min} = \Delta V_{max} + (n-1)\Delta V_S \tag{26}$$

and goes up to the asterisk marker at

$$A_{max} = \Delta V_{max} + n\Delta V_S \tag{27}$$

if there is another tap position above the current position. If the current tap position is the maximum tap position, A_{max} goes to infinity and the inverse describing functions goes to $N_I(A) = -1$. A more detailed analysis and proof on all of the above can be found in [13].

The validation of the describing function is done by comparing the expected oscillations with the oscillations of a detailed RDT-model with an artificial linear transfer function (28), which fulfills all requirements for the Harmonic Balance and has intersections with the RDT's describing functions.

$$G(s) = \frac{0.04(s^3 + 30s^2 - 10s + 1)}{(0.5s+1)^6 + 2} \tag{28}$$

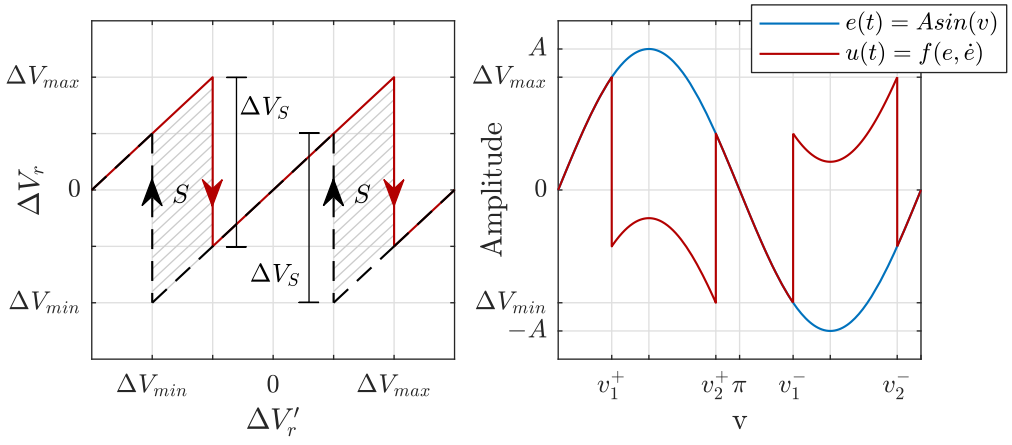


Fig. 5. Characteristic of a RDT (left) without time delay (dotted for $\dot{\Delta}V'_r > 0$; solid for $\dot{\Delta}V'_r < 0$) and the respective oscillation (right).

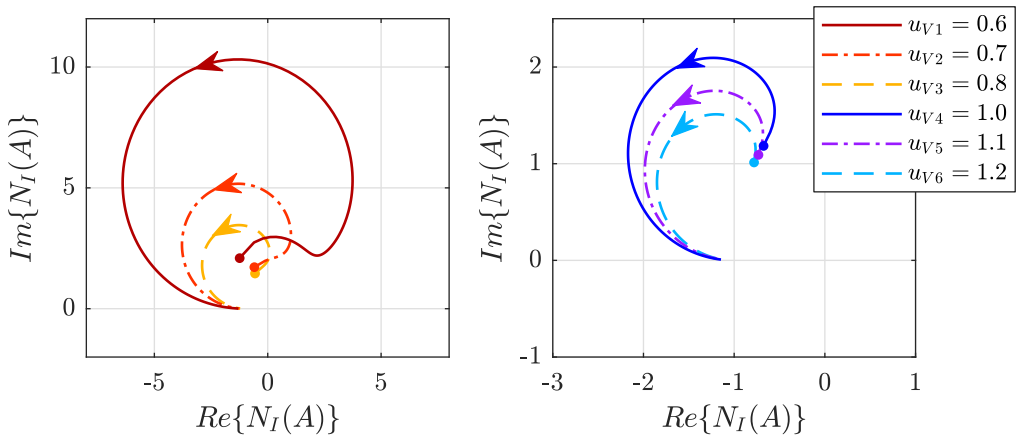


Fig. 6. $N_I(A)$ of an RDT for different u_v.

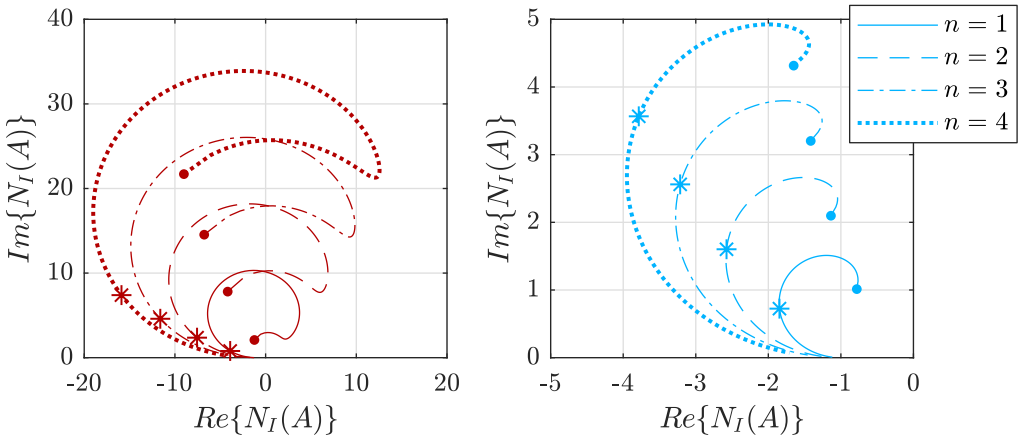


Fig. 7. $N_I(A)$ of an RDT with for n tap positions for $u_v = 0.6$ (left) and $u_v = 1.2$ (right).

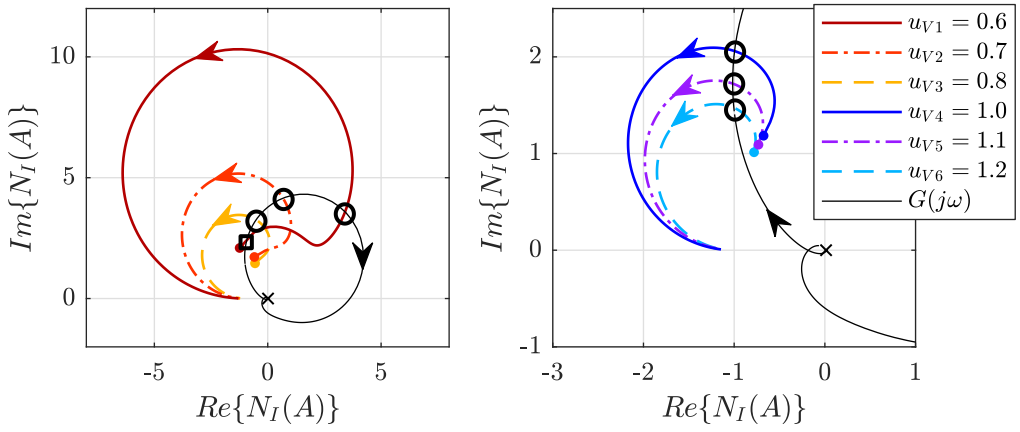


Fig. 8. Intersections of $N_I(A)$ and $G(s)$ for different u_v; circle: stable OP, square: unstable OP.

Table 1
Utilized voltages for further testing.

u_{vi}	$\Delta V_S inpu$	$\Delta V_{max} inpu$	u_v
u_{v1}	0.025	0.015	0.6
u_{v2}	0.025	0.0175	0.7
u_{v3}	0.025	0.02	0.8
u_{v4}	0.025	0.025	1.0
u_{v5}	0.025	0.0275	1.1
u_{v6}	0.025	0.03	1.2

By plotting both and $G(s)$ in the complex plane, as seen in Fig. 8 for different u_v, the frequency and the amplitude of the stable oscillations can be graphically determined. The intersection marked with a square is the only unstable operating point (OP) due to the relative direction of the graphs to each other. The intersections marked with a circle are the expected operating points, where an oscillation can persist.

The simulations of the control system with the nonlinear characteristic and the linear transfer function show, that after a short impulse, stable oscillations occur with minute differences to the expected amplitude and frequency, as listed in Table 2 for the u_{vi} in Table 1.

The plots of the oscillations for u_{v1} and u_{v6} in Fig. 9 show, that the

Table 2
Validation of the graphically determined (HB) oscillations with simulated (numerical model) oscillation variables of an RDT without time delay and a controller with the artificial transfer function $G(s)$.

u_V	A_{graph} in pu	ω_{graph} in $\frac{rad}{s}$	A_{sim} in pu	ω_{sim} in $\frac{rad}{s}$	ε_A	ε_ω
0.6	0.0151	1.048	unstable OP	unstable OP	–	–
	0.0233	1.126	0.0232	1.131	−0.4%	0.4%
0.7	0.0260	1.093	0.0259	1.095	−0.4%	0.2%
0.8	0.0282	1.068	0.0282	1.069	0%	0.1%
1.0	0.0318	1.031	0.0319	1.030	0.3%	0.1%
1.1	0.0333	1.016	0.0333	1.015	0%	−0.1%
1.2	0.0347	1.002	0.0345	1.002	−0.6%	−0.1%

fundamental wave, which represents the graphical determined oscillation from Table 2, and the simulated oscillation of $\Delta V_r'$ and ΔV_r are almost identical during steady-state operation. However, variations of $G(s)$ show, that the error increases for linear transfer functions with a weaker low-pass filter characteristic due to the increased influence of the neglected high order harmonics. A stable oscillation may still occur, however, the analytically determined values do not match the simulated values in that case [13].

3.3. RDT-model with time delay

In order to evaluate the effect of an RDT's time delay on $N(A)$, a time dependent characteristic can to be defined as depicted in Fig. 10. The resulting switching points v' are defined in Eq. (29), where ω is the angular velocity of the oscillation and t_v is the time delay of the RDT.

$$v' = v + \omega t_v \tag{29}$$

The boundary voltages are time dependent as well and can be defined by Eqs. (30) and (31). The inverse describing function $N_I(A, \omega)$ is calculated by piecewise integrating over the characteristic.

$$\begin{aligned} \Delta V_{max,t} &= A\sin(v_1'^{+}) \\ &= A\sin(v_1^{+} + \omega t_v) \end{aligned} \tag{30}$$

$$\begin{aligned} \Delta V_{min,t} &= A\sin(v_2'^{+}) - \Delta V_S \\ &= A\sin(v_2^{+} + \omega t_v) - \Delta V_S \end{aligned} \tag{31}$$

As one period of an RDT with ± 1 tap positions is comprised of four switching actions, a tap position change must occur while $v_{max} \leq \frac{\pi}{2}$. Thus, a condition for a stable oscillation can be set, each depending on the time delay t_v. The minimum duration T_{min} and the maximal angular velocity ω_{max} are defined by (32).

$$\begin{aligned} T_{min} &\geq 4t_v \\ \omega_{max} &\leq \frac{\pi}{2t_v} \end{aligned} \tag{32}$$

The minimum amplitude for an RDT without time delay is defined as $A \geq \Delta V_{max}$, whereas the minimum amplitude A_{min} has to be adapted to (33) due to the shifted voltage boundary in the characteristic in Fig. 10.

$$A_{min} \geq \frac{\Delta V_{max}}{\sin(\frac{\pi}{2} - \omega t_v)} \tag{33}$$

In order to graphically analyze the intersections of the linear transfer function and the describing function, a 3D-plot is necessary. The resulting graphs for two exemplary u_v are depicted in Fig. 11. The plots remain in the positive imaginary half-plane and approach $N_I = -1$ for $A \to \infty$. The intersections at A_{graph} and ω_{graph} of the describing function and the linear transfer function can be found in Table 3.

The comparison between the graphically determined oscillation variables from Fig. 11 and the simulation results in Table 3 and Fig. 12 show, that the values deviate only marginally from each other. Thus, this method allows to predict oscillations for an RDT with a time delay.

Different time delays, as shown in Fig. 13, only affect the steepness of the describing function as can be seen by the y-axis scaling. The projection of the plot onto the complex plane shows no differences. A stable oscillation therefore heavily depends on the time delay t_v and the graph of $G(s)$.

3.4. Critical area

The previous results lead to two criteria, which can be applied to linear transfer functions to rule out oscillations due to harmonic balance between an RDT and a linear controller. The first criteria is based

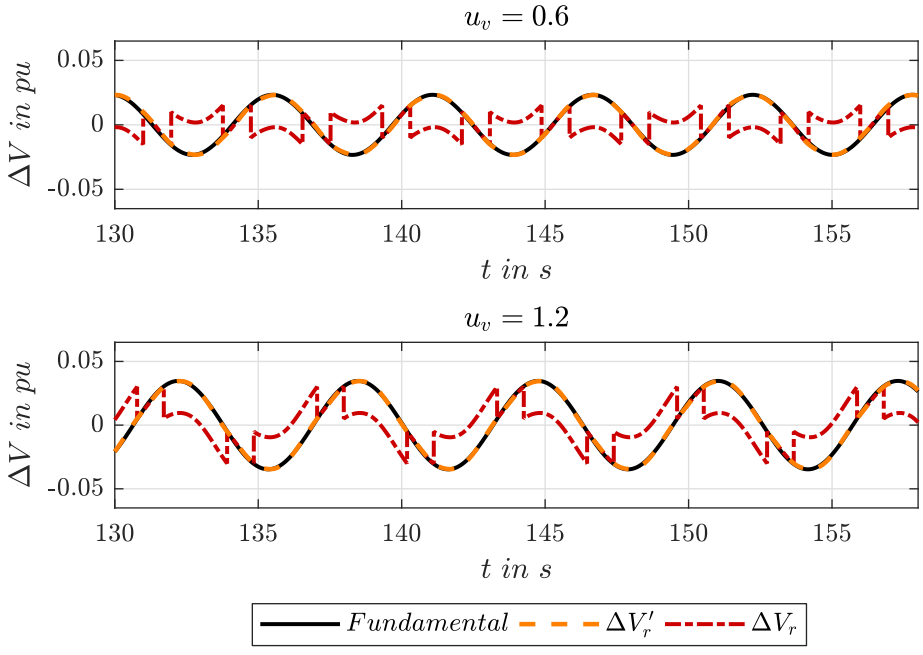


Fig. 9. Oscillation in a simulation of an RDT without time delay and a controller with the artificial transfer function $G(s)$.

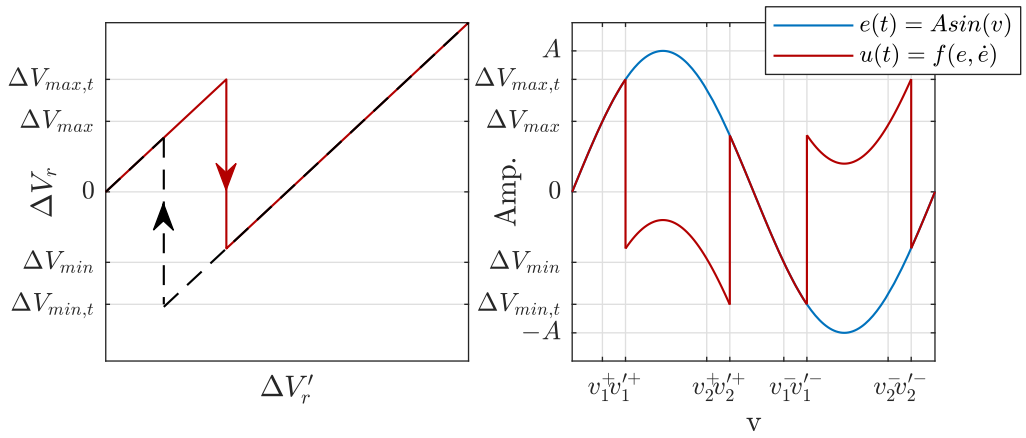


Fig. 10. Characteristic (left) of a time dependent RDT (dotted for $\Delta\dot{V}_r' > 0$; solid for $\Delta\dot{V}_r' < 0$) and the respective oscillation (right).

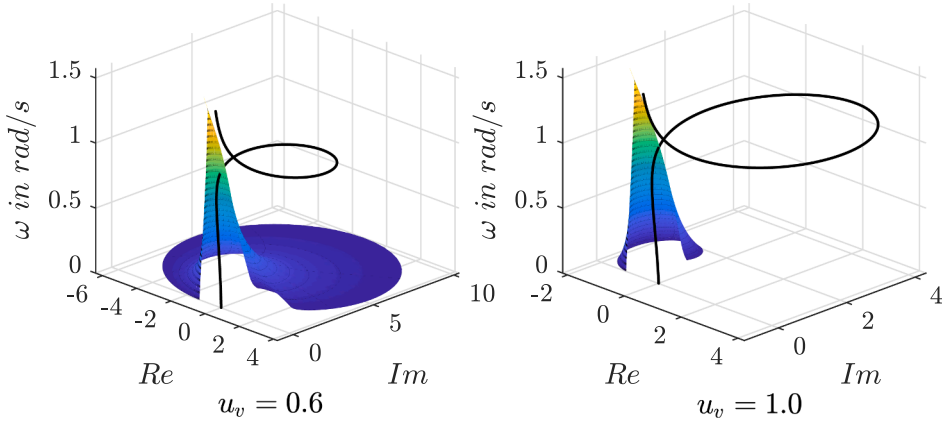


Fig. 11. Intersections of $G(s)$ with $N_I(A, \omega)$ for different u_v with $t_v = 1$.

Table 3
Validation of the graphically determined (HB) oscillations with simulated (numerical model) oscillation variables of an RDT with time delay $t_v = 1s$ and a controller with the artificial transfer function $G(s)$.

u_V	A_{graph} in pu	ω_{graph} in $\frac{rad}{s}$	A_{sim} in pu	ω_{sim} in $\frac{rad}{s}$	ε_A	ε_ω
0.6	0.0430	0.943	0.0412	0.927	−4.2%	−1.7%
1.0	0.0441	0.923	0.0410	0.906	−6.6%	−1.8%

on the observation, that the imaginary part of $N_I(A, \omega)$ is always greater zero. Thus, the first criteria is defined as follows.

1. **Criteria:** If $Im\{G(s)\} \leq 0$ for all $s \in [0, \infty[$, oscillations due to harmonic balance can not occur.

   The area of the complex plane, where an intersection of a linear transfer function with the describing function can only occur, is defined by the projection of the time dependent graph of $N(A, \omega)$ onto the complex plane. The graph of $N_I(A, \omega)$ for $u_v \to 0.5$ shows the largest span of any describing function. Thus, it is used to define the minimum area in the positive imaginary half plane, where an intersection is not possible. This area is shown in Fig. 14.

   The save area can be approximated by the area of the two half circles with the centers $c_1 = (0|1.617)$ and $c_2 = (0| - 0.2)$ and the radii $r_1 = 2.058$ and $r_2 = 0.8$. An oscillation due to harmonic balance can occur, if the graph of the transfer function runs through the area above the imaginary axis with the exception of the area between the half circles and the imaginary axis. Thus the second criteria can be defined as follows:

2. **Criteria:** For linear transfer functions, whose locus does not cross into the imaginary half plane with exception of the two half circles with the centers $c_1 = (0|1.617)$ and $c_2 = (0| - 0.2)$ and the radii $r_1 = 2.058$ and $r_2 = 0.8$, an oscillation due to harmonic balance can not occur

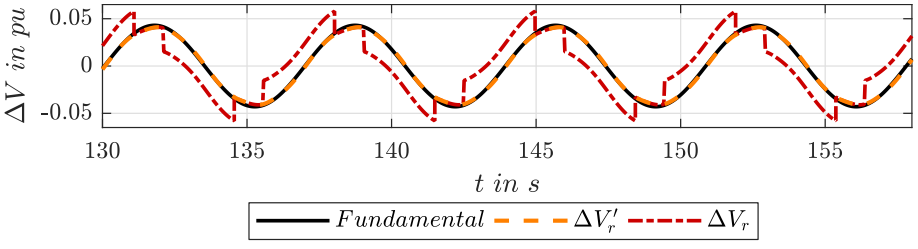


Fig. 12. Oscillation in a simulation of an RDT with $u_v = 1.0$ and time delay $t_v = 1s$ and a controller with the artificial transfer function $G(s)$.

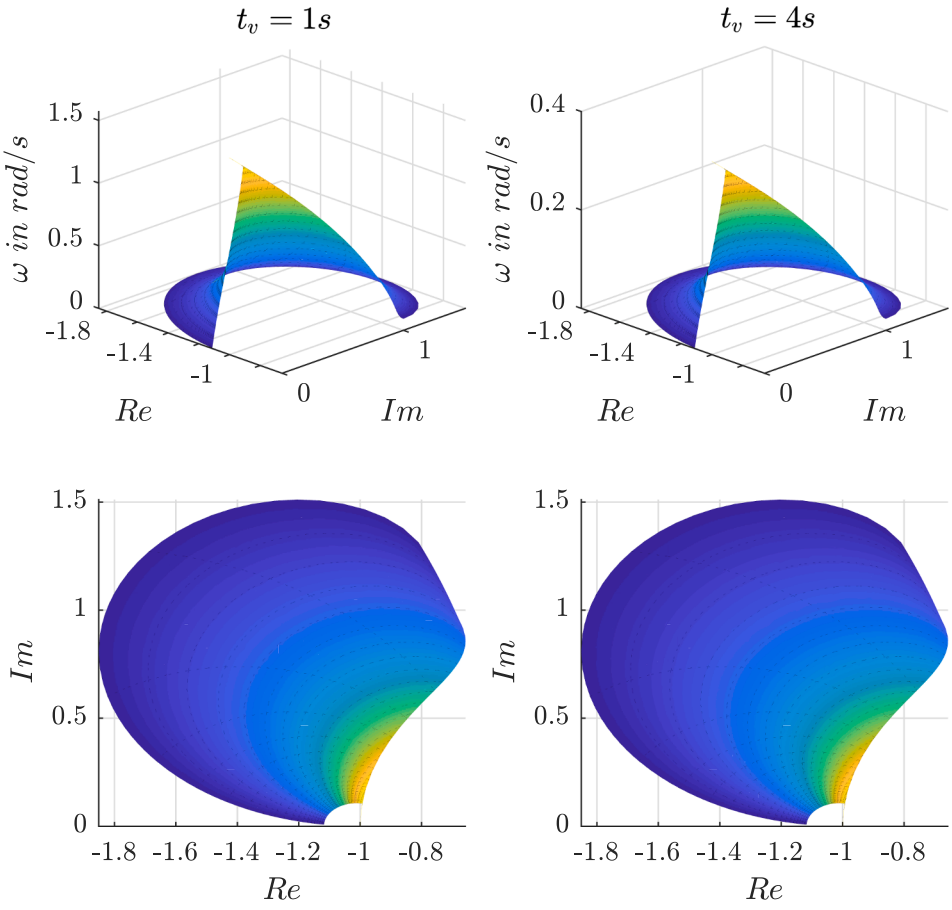


Fig. 13. $N_I(A, \omega)$ for different t_v with $u_v = 0.6$.

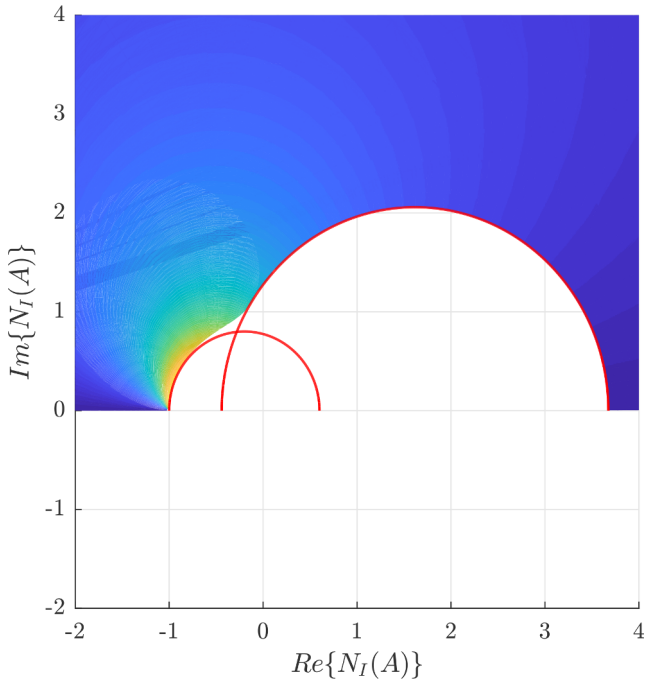


Fig. 14. Critical area defined by $N_I(A, \omega)$ for $u_v \to 0.5$.

A violation of one of the criteria does not necessarily lead to oscillations, since the actual describing function may not intersect with the locus due to the angular velocity and RDT parametrization. Therefore, both criteria are sufficient but not necessary to rule out oscillations. A detailed analysis with the actual describing function of the RDT and the linear transfer function of the controller is necessary if the second criteria is violated.

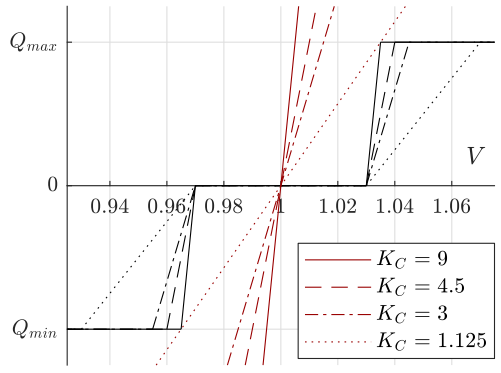


Fig. 15. Simplified characteristics of the Q(V)-control.

4. Interaction with local voltage control

On the basis of the two defined criteria, two real inverters (A and B) are analyzed and tested for the occurrence of oscillations with an RDT according to the harmonic balance. Both controllers are simplified in regards to their non-linear characteristic, which is replaced by a gain equal to the maximum gradient of the actual characteristic, as shown in Fig. 15 for different gains. Since the characteristics limit the maximum reactive power and thus the maximum amplitude of the voltage influence any stable operating point without the characteristic can be assumed to be stable if the limiter is applied.

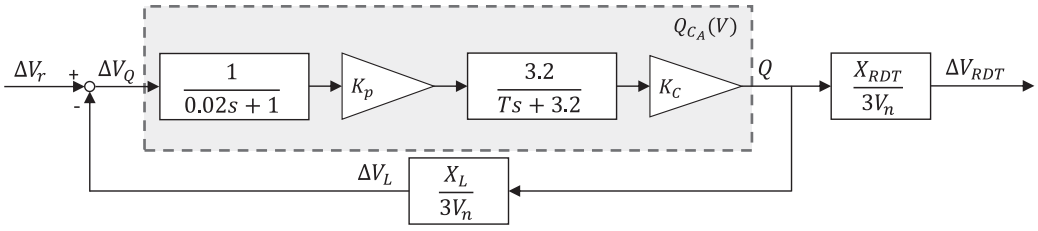


Fig. 16. Simplified model of the controller A and the grid.

Table 4
Reference parameterization for the controller A (black graphs in Fig. 17).

K_C	T	X_L	X_{RDT}
9	1	1.031 pu	0.01855 pu

Furthermore, gradient limiters are not taken into consideration, as the maximum gradient of an oscillation usually falls below the threshold. Thus, the simplified model can be inserted into the control model in Fig. 4 and combined to a linear transfer function.

4.1. Controller A

The simplified model of the Q(V)-control A is based on [14] and the respective feedback loop of the grid is depicted in Fig. 16, which can be summarized to the linear transfer function (34) for $V_n = 1$.

$$G_{C_A}(s) = \frac{K_C K X_{RDT}}{0.01875Ts^2 + (0.06 + 0.9375T)s + K_C K X_L + 3} \tag{34}$$

By evaluating the locus of (34) for different parametrization with the parameters in Table 4 as reference parameters, the general behaviour of $G_{C_A}(s)$ in Fig. 17 is analyzed. The graph of each locus stays in the negative imaginary half plane. Therefore, according to criteria 1, an oscillation due to the harmonic balance of the RDT and the Q(V)-control can not occur.

4.2. Controller B

The model of controller B in Fig. 18 can be simplified to the linear transfer function (36) with (35).

$$Q_{C_B}(s) = \frac{KK_C}{(0.7s+1)(0.04s+1)(0.02s+1)(0.2Ts+1)} \tag{35}$$

$$G_{C_B}(s) = \frac{X_{RDT}}{3V_n} \frac{Q_{C_B}(s)}{1 + \frac{X_L}{3V_n} Q_{C_B}(s)} \tag{36}$$

Fig. 19 shows three different scenarios. The reference locus (black) is defined by the reference X_{RDT} and X_L and $K_C = 9$, $K = 29$, which represent high gains. Since it doesn't violate the second criteria, an oscillation due to harmonic balance can not occur. For the second locus (green), the Q(V) control is connected directly at the RDT, thus, $X_L = 0$. The result is a locus, which satisfies the first criteria. For the third locus (orange), X_{RDT} was doubled. The graph violates the second criteria. However, the locus encompasses the point $(-1|0)$ thus violating the Nyquist-criteria. The resulting Q(V)-controller itself is therefore

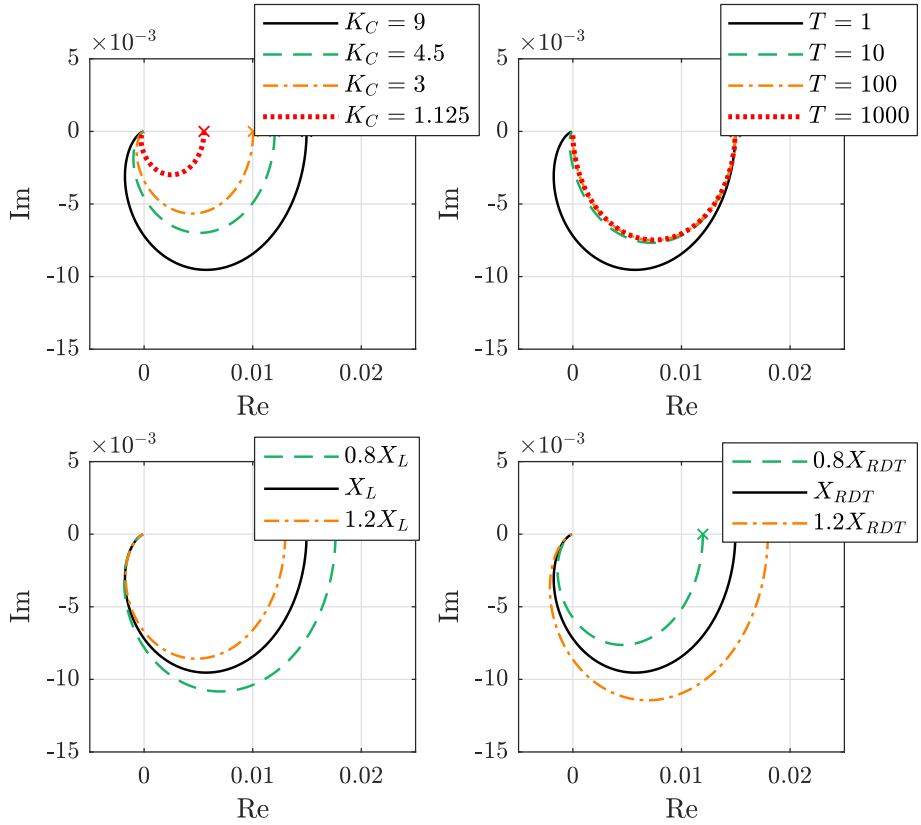


Fig. 17. Locus of $G_{C_A}(s)$ for different parameters (from marker x at $s = 0$ to $s \to \infty$).

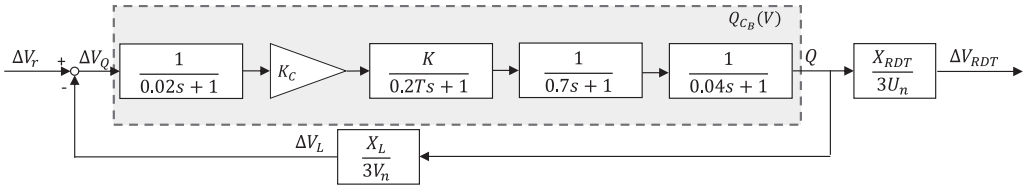


Fig. 18. Simplified model of the controller A and the grid.

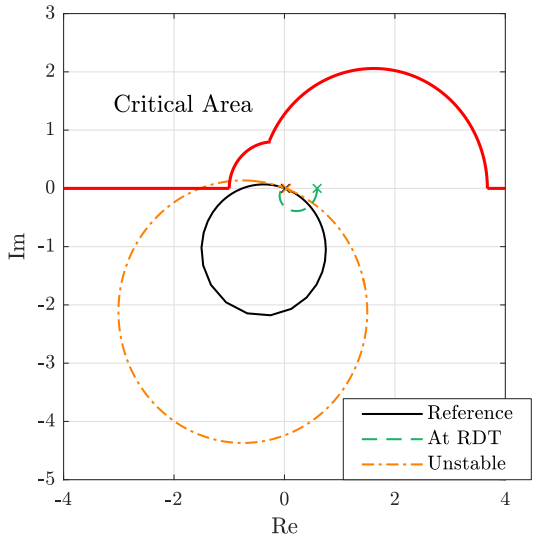


Fig. 19. Locus of $G_{C_B}(s)$ for $K_C = 9$, $K = 29$ with reference X_{RDT} and X_L (black), $X_L = 0$ (green) and $2X_{RDT}$ (orange). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

unstable. A oscillation may still occur, as non-linear switching controllers like an RDT can stabilize unstable linear controllers.

5. Conclusion

The harmonic balance method has been shown to be a valid method to analyze interactions of a RDT with time delay and a Q(V) controller. The two criteria in Section 3.4 for the locus of a linear transfer function (i.e. Q(V)-control) allow to predict the occurrence of oscillations in electrical grids. A detailed analysis of the RDT's describing function and the linear function is necessary for critical combinations where a violation of the criteria occurs. A safe operation without the risk of an oscillation may still be possible depending on the frequency behavior of the describing function.

Furthermore, oscillations due to the harmonic balance for the examined Q(V)-control systems has been ruled out based on the defined criteria. An additional analysis concerning the overall system stability of the control system is still necessary, since the harmonic balance method allows no conclusions in that regard. Furthermore, the influence of the nonlinear behaviour of the Q(V)-control should be further investigated, as they may stabilize the system even further beyond the defined criteria. An analysis examining the control structure of the Q(V)-control leading to a critical locus would allow a simplified way to identify critical controller combinations with an RDT without detailed simulations.

CRediT authorship contribution statement

T. Würl: Conceptualization, Methodology, Software, Validation, Writing - original draft, Writing - review & editing, Visualization. **M. Lindner:** Conceptualization, Methodology, Writing - review & editing. **R. Witzmann:** Supervision, Resources.

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